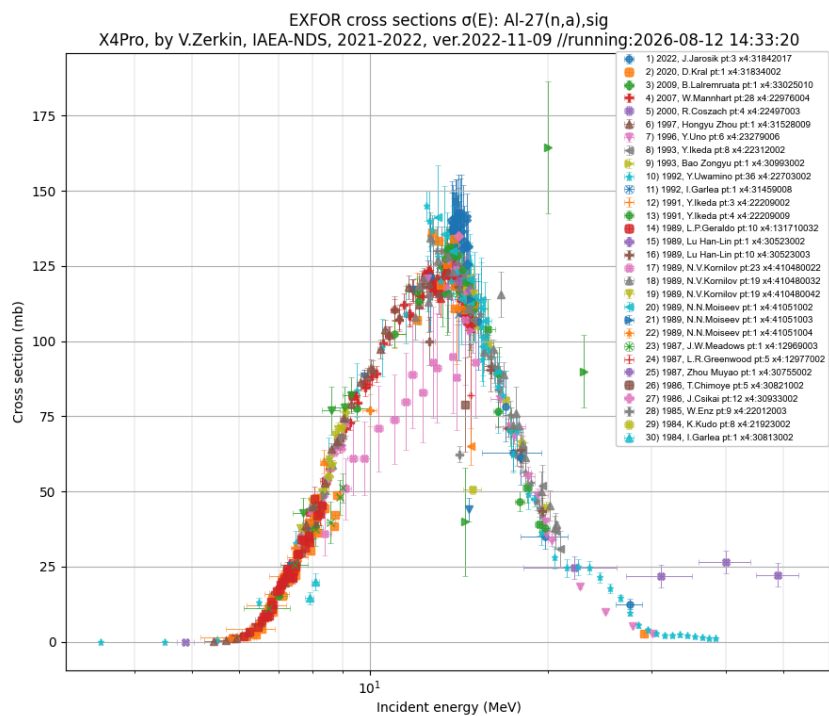


X4Pro. Plotting via Matplotlib.

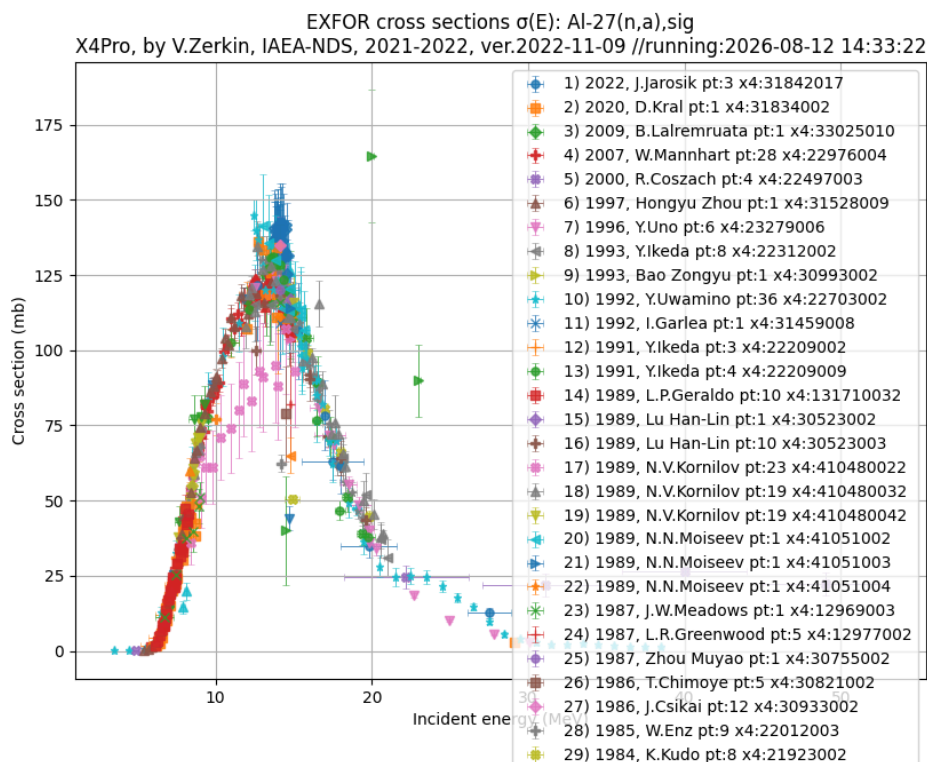
Generated: 2026-08-14,16:24:07

part1-1-sig/ Retrieve and plot EXFOR cross sections (modular Python-program)

1) part1-1-sig/out00/Al27na-1.png

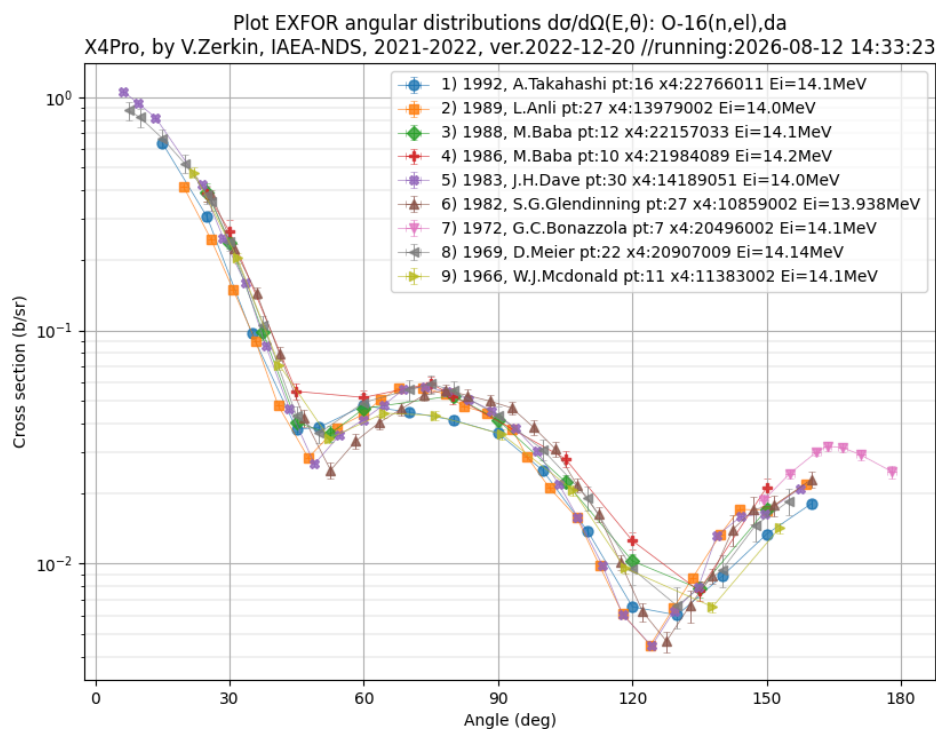


2) part1-1-sig/out00/Al27na-2.png



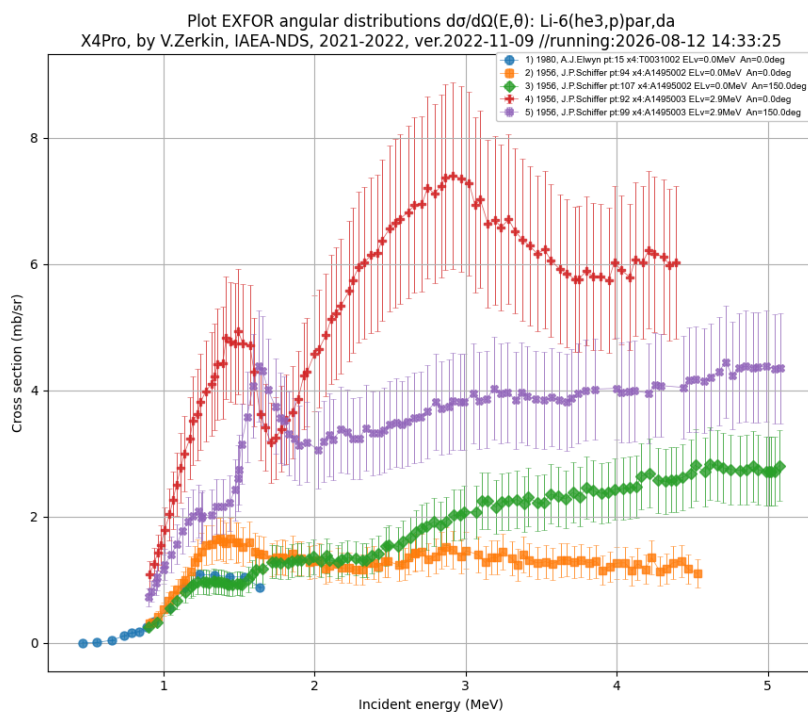
part1-2-da/ EXFOR angular distributions

3) part1-2-da/out00/da0an.png



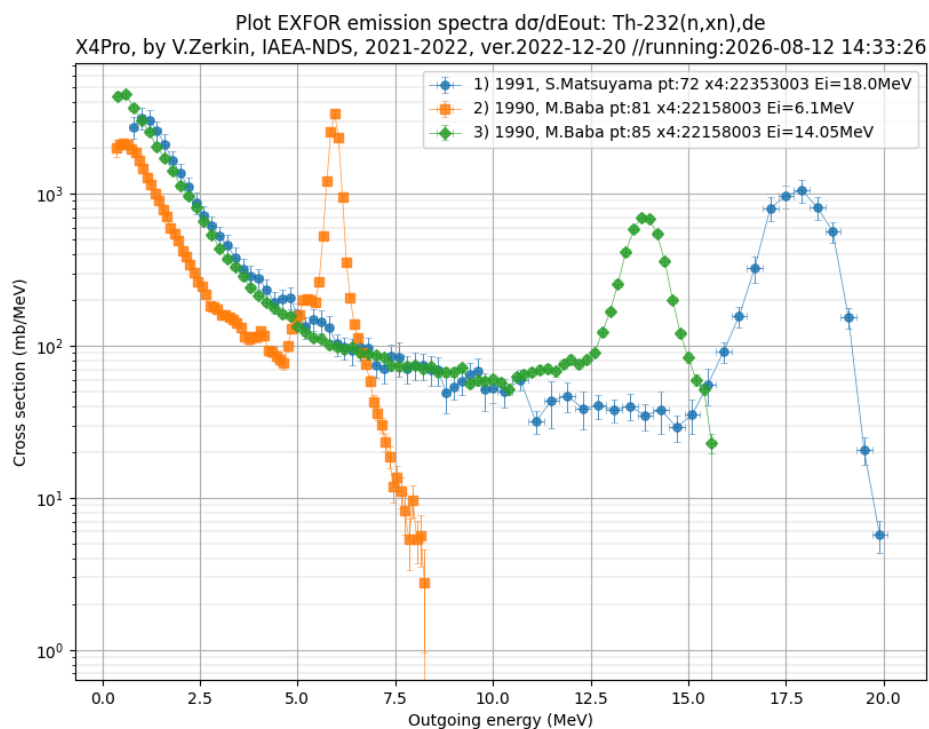
part1-3-dap/ EXFOR partial angular distributions

4) part1-3-dap/out00/dap0ei.png



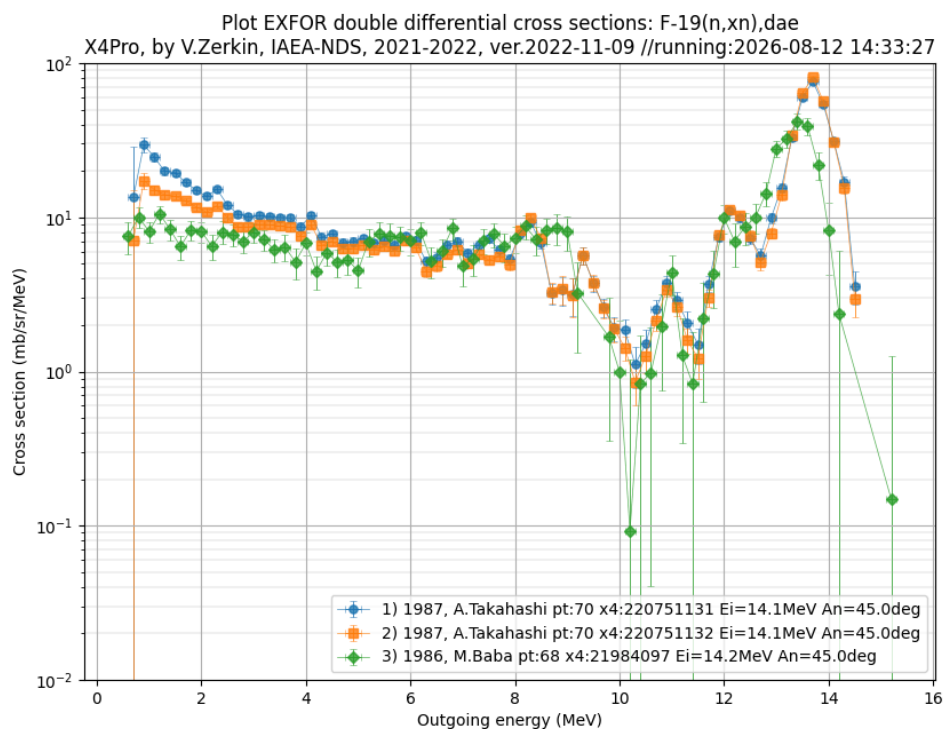
part1-4-de/ EXFOR energy spectra of outgoing particles

5) part1-4-de/out00/de0eo.png



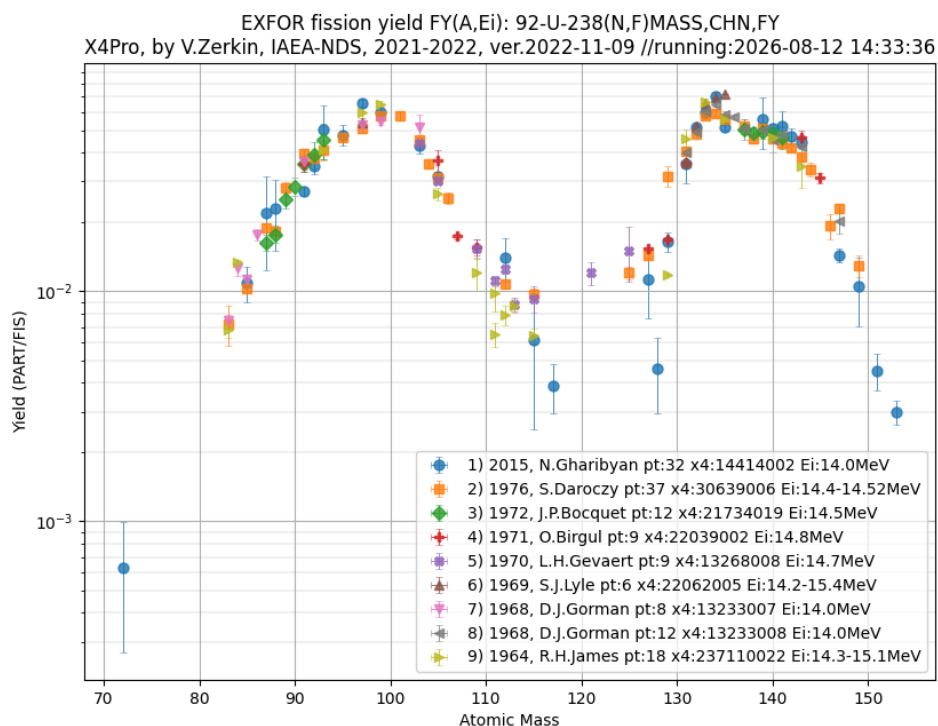
part1-5-dae/ EXFOR double differential cross sections

6) part1-5-dae/out00/dae0eo.png

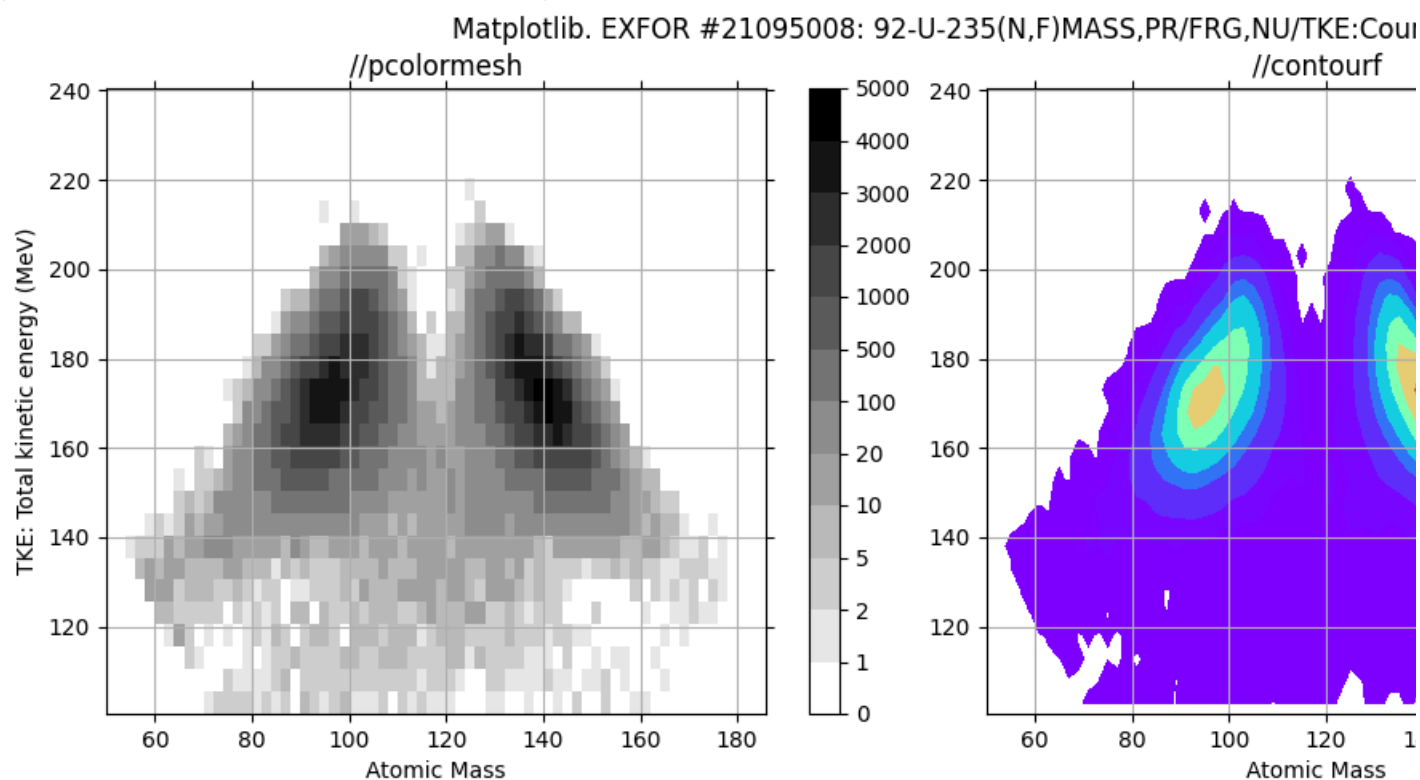


part1-6-fy/ EXFOR fission yield

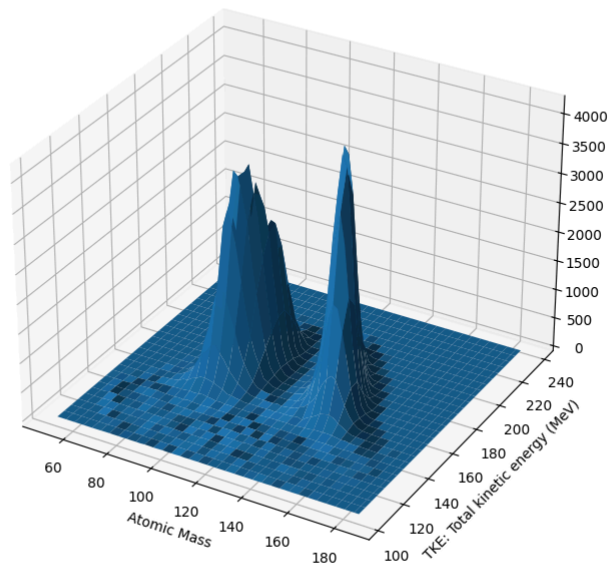
7) part1-6-fy/out00/U238nf-fy.png



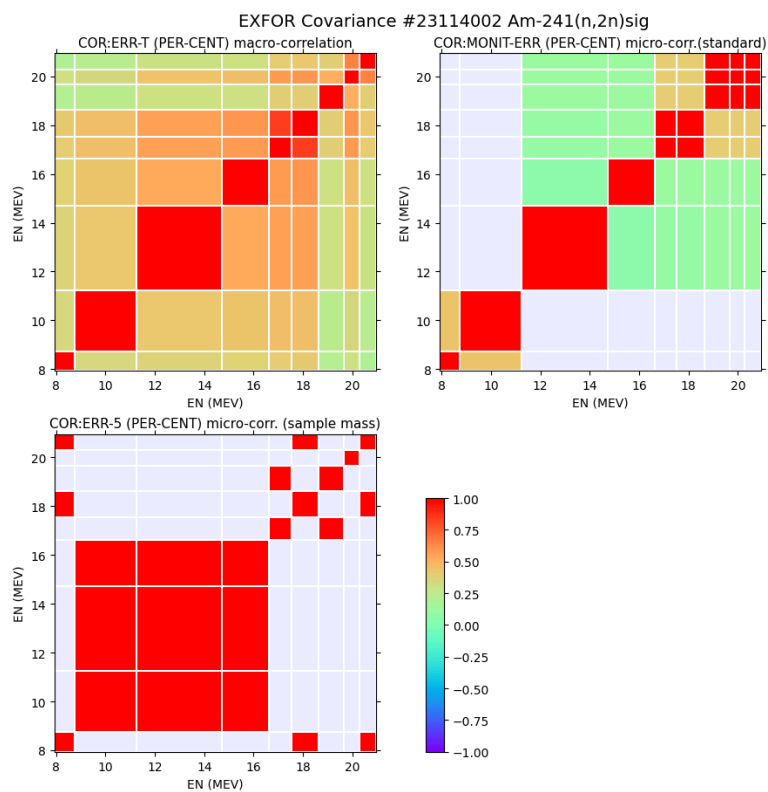
8) part1-6-fy/out00/me0mpl2d2.png



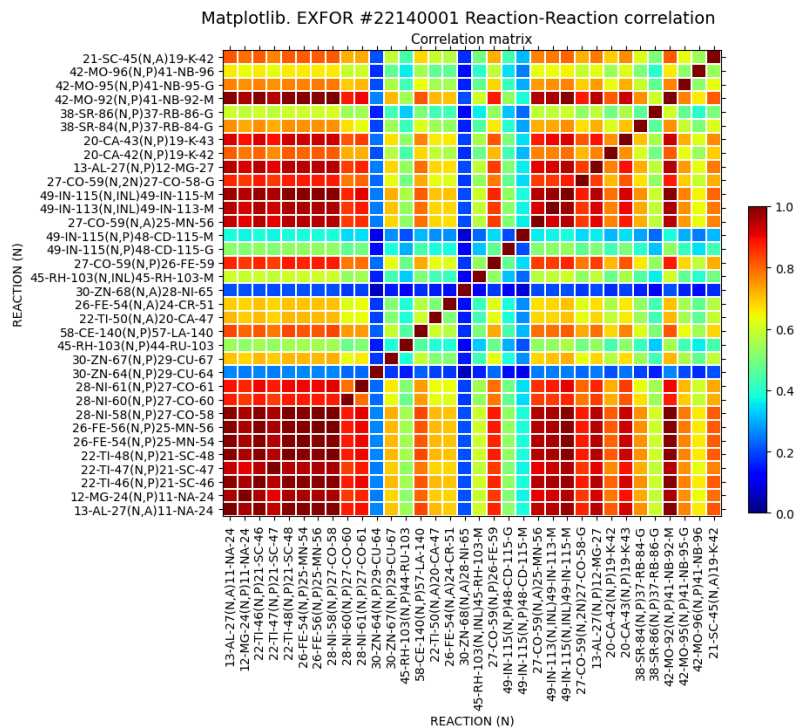
Matplotlib. EXFOR #21095008: 92-U-235(N,F)MASS,PR/FRG,NU/TKE:Counts



10) part1-7-covar/out00/covar1.png

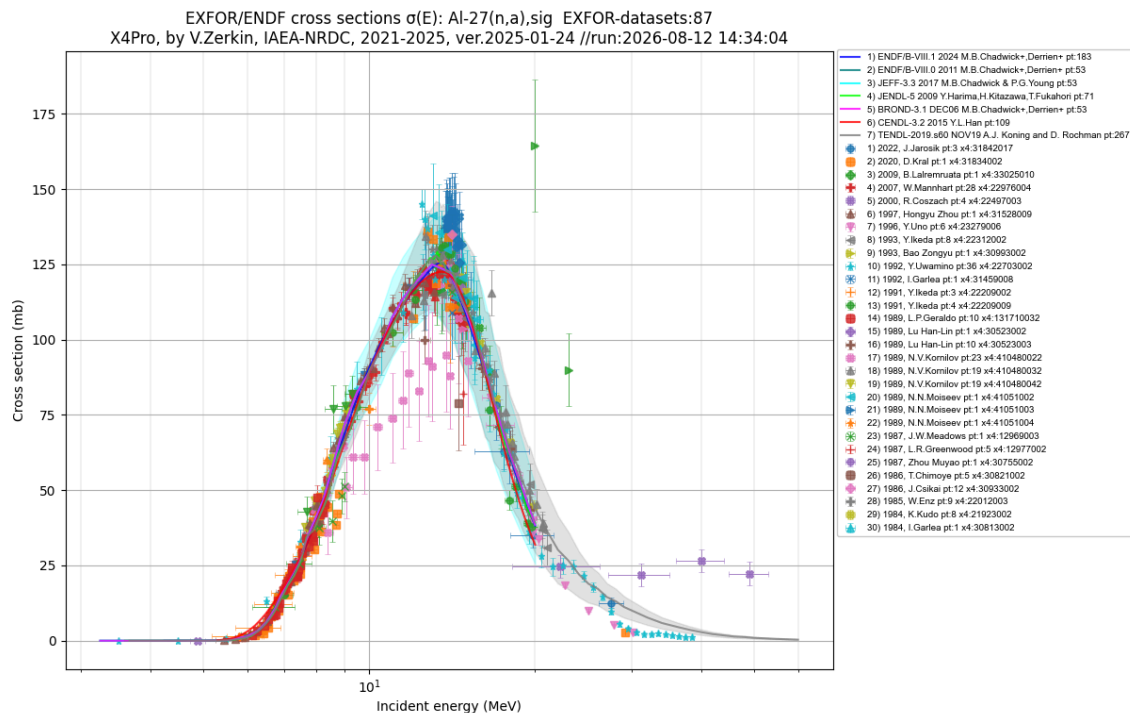


11) part1-7-covar/out00/covar2.png

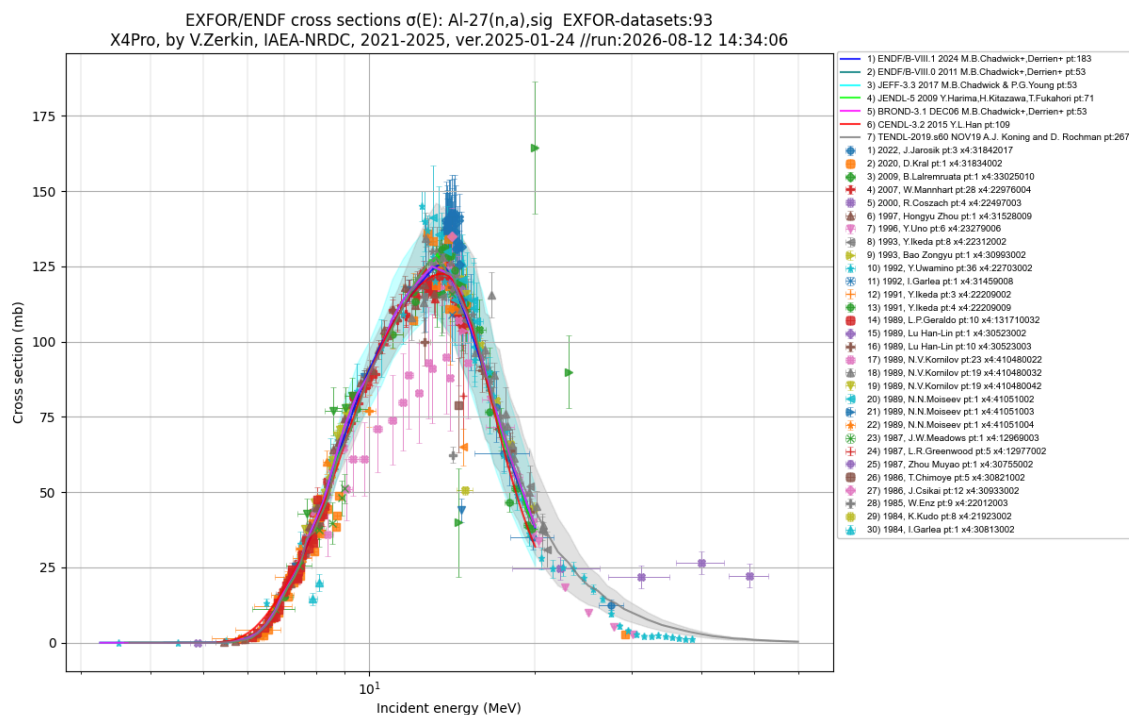


part2-1-sig1/ Retrieve and plot EXFOR/ENDF cross sections

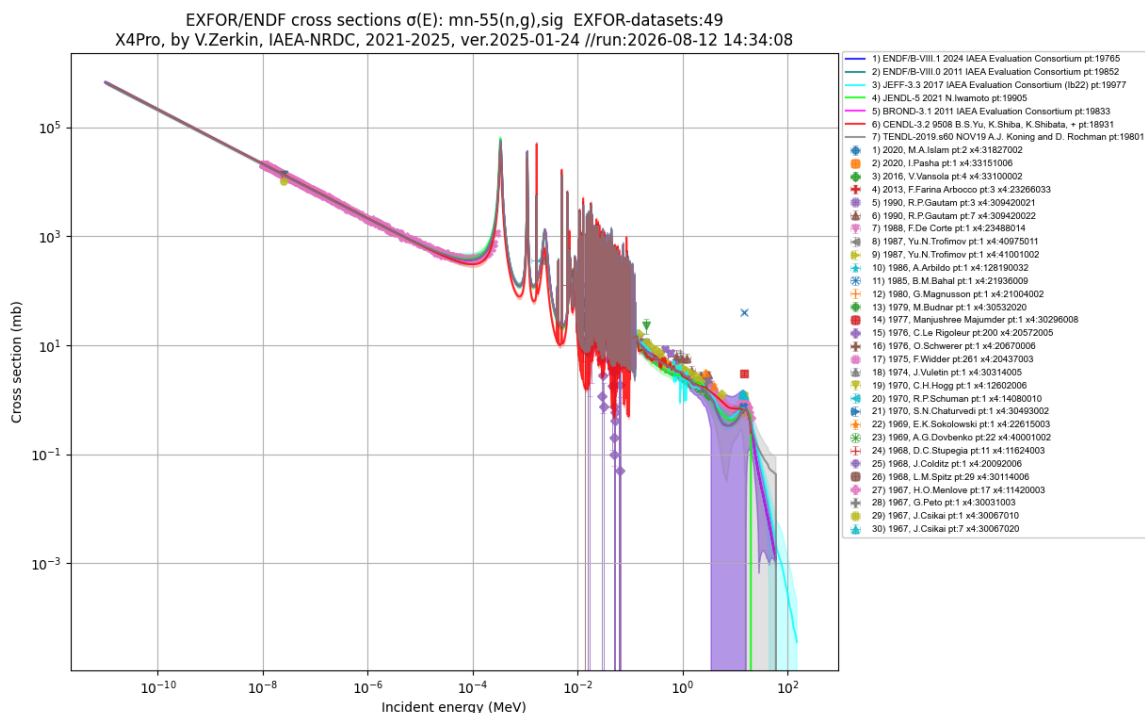
12) part2-1-sig1/out00/Al27na.png



13) part2-1-sig1/out00/Al27na2.png

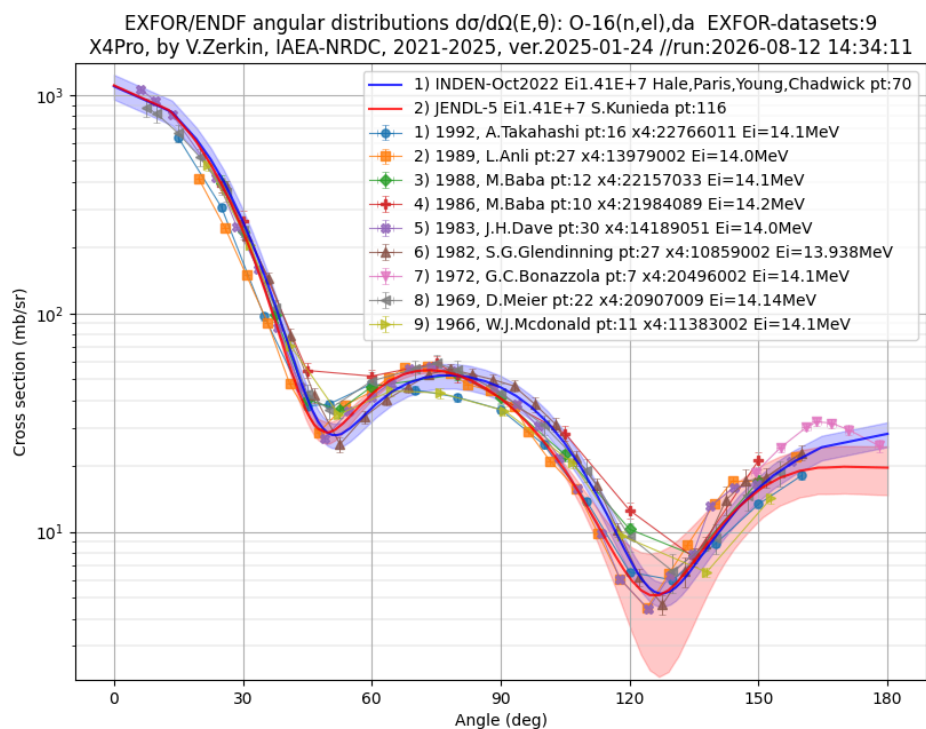


14) part2-1-sig1/out00/mn55ng.png



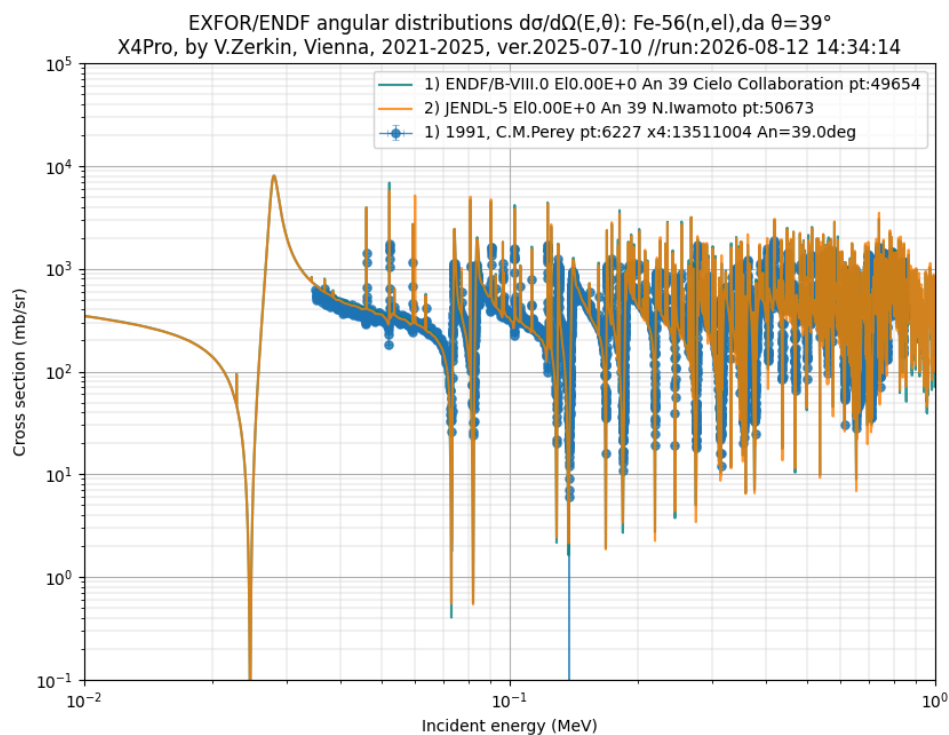
part2-2-da1an/ EXFOR/ENDF angular distributions (Ei:fixed)

15) part2-2-da1an/out00/da1an.png

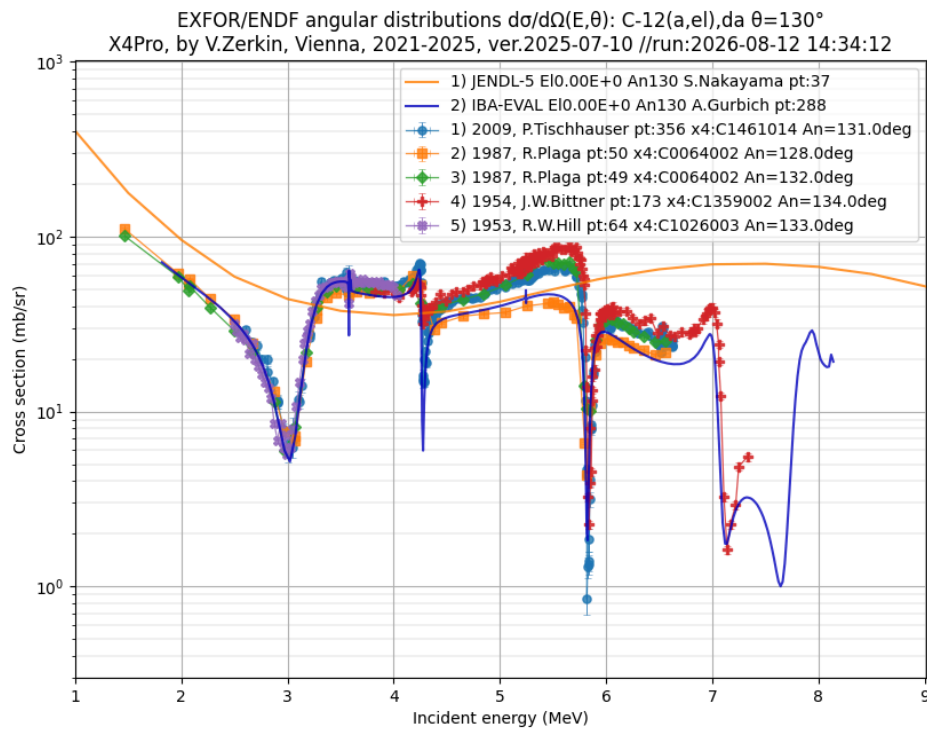


part2-3-da1ei/ EXFOR/ENDF angular distributions (angle:fixed)

16) part2-3-da1ei/out00/da1ei-ex2.png

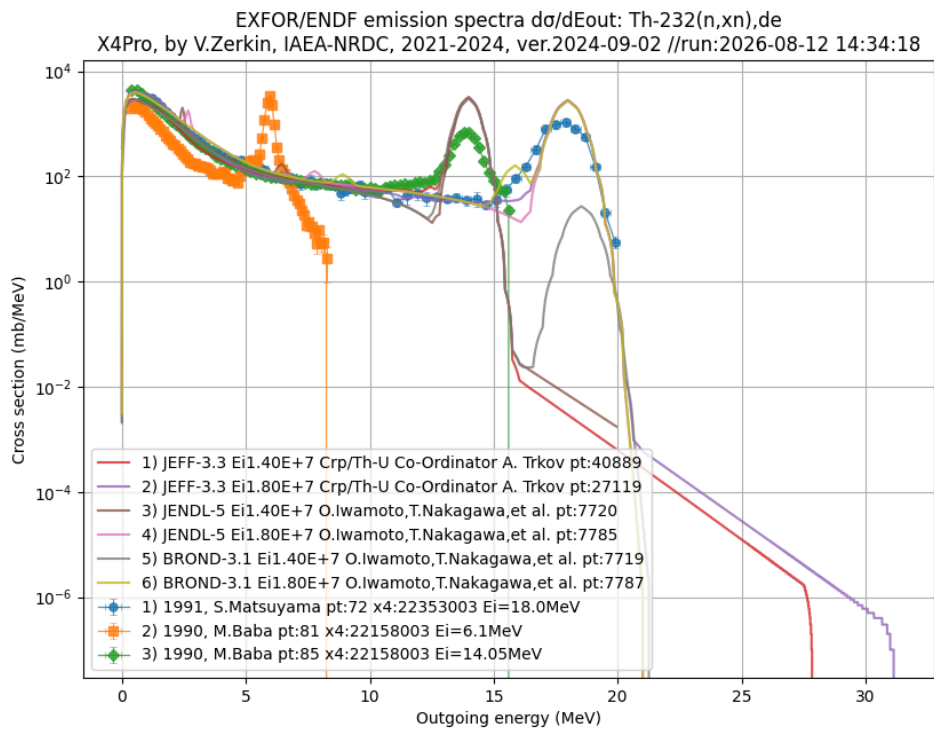


17) part2-3-da1ei/out00/da1ei.png



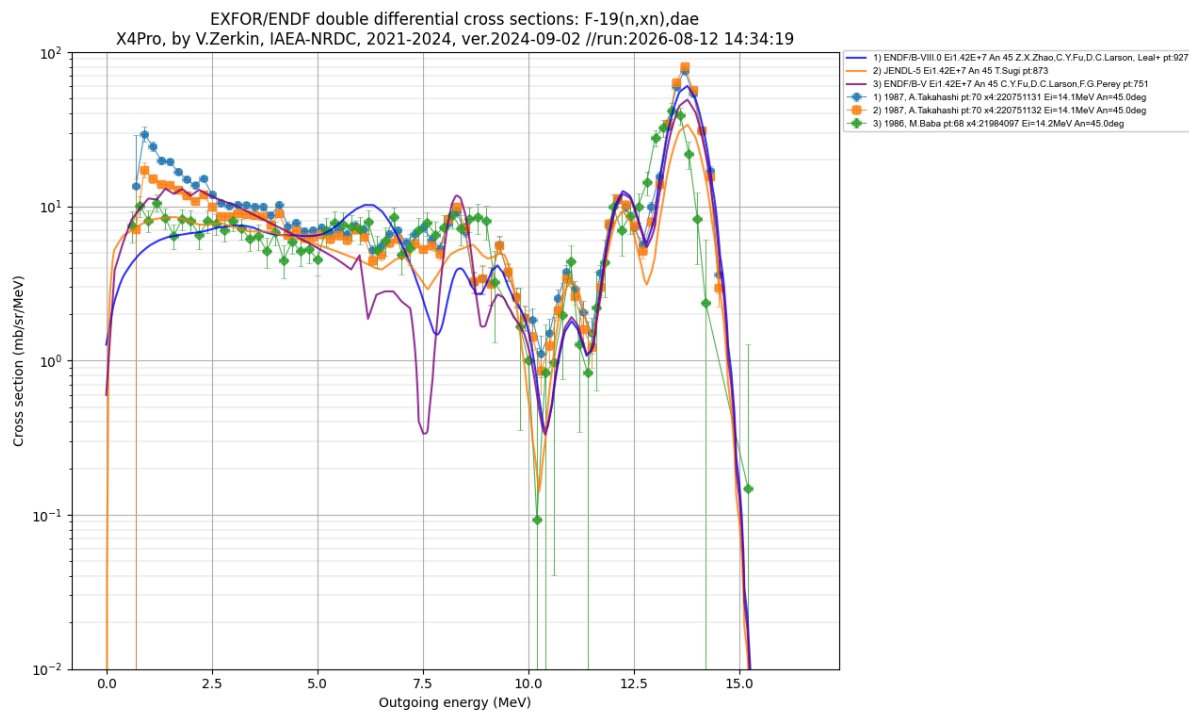
part2-4-de1eo/ EXFOR/ENDF energy spectra of outgoing particles

18) part2-4-de1eo/out00/de1eo.png



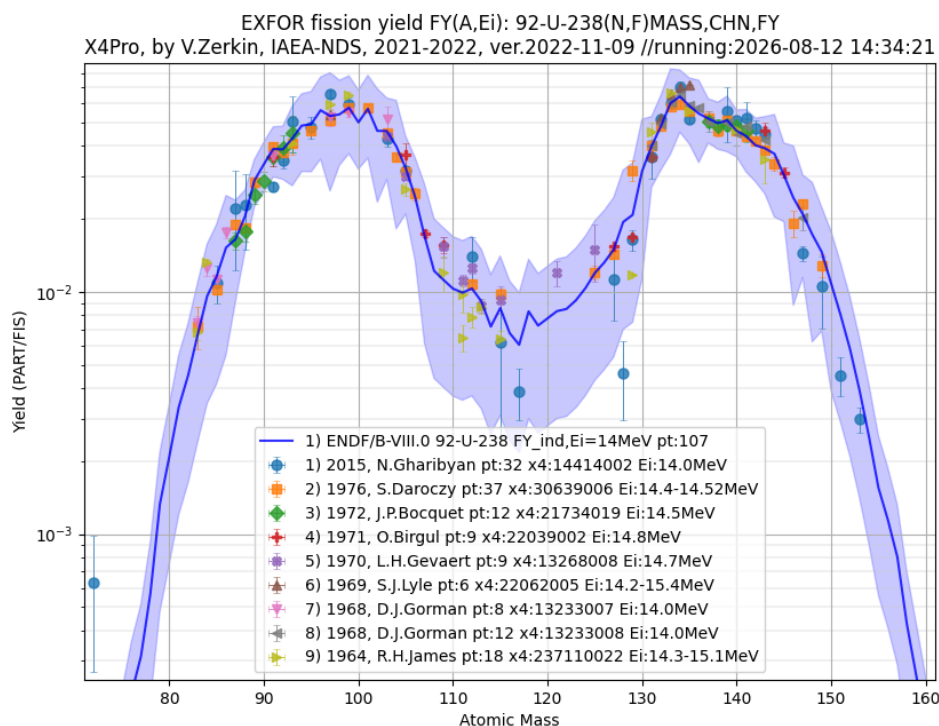
part2-5-dae1eo/ EXFOR/ENDF double differential cross sections

19) part2-5-dae1eo/out00/dae1eo.png



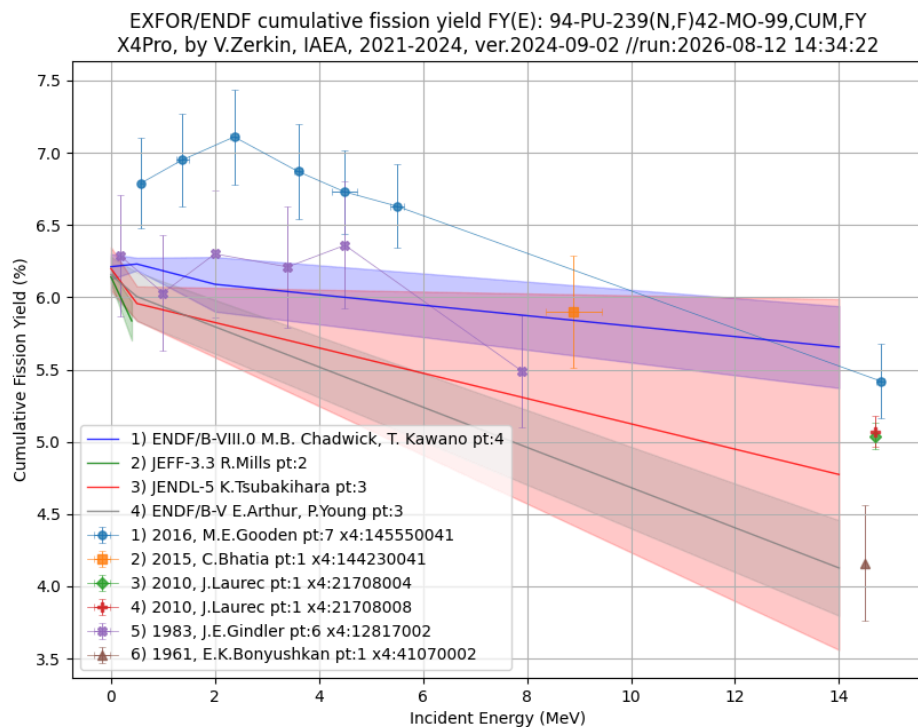
part2-6-fy/ EXFOR/ENDF fission yeild (mass distribution)

20) part2-6-fy/out00/U238nf-fy.png

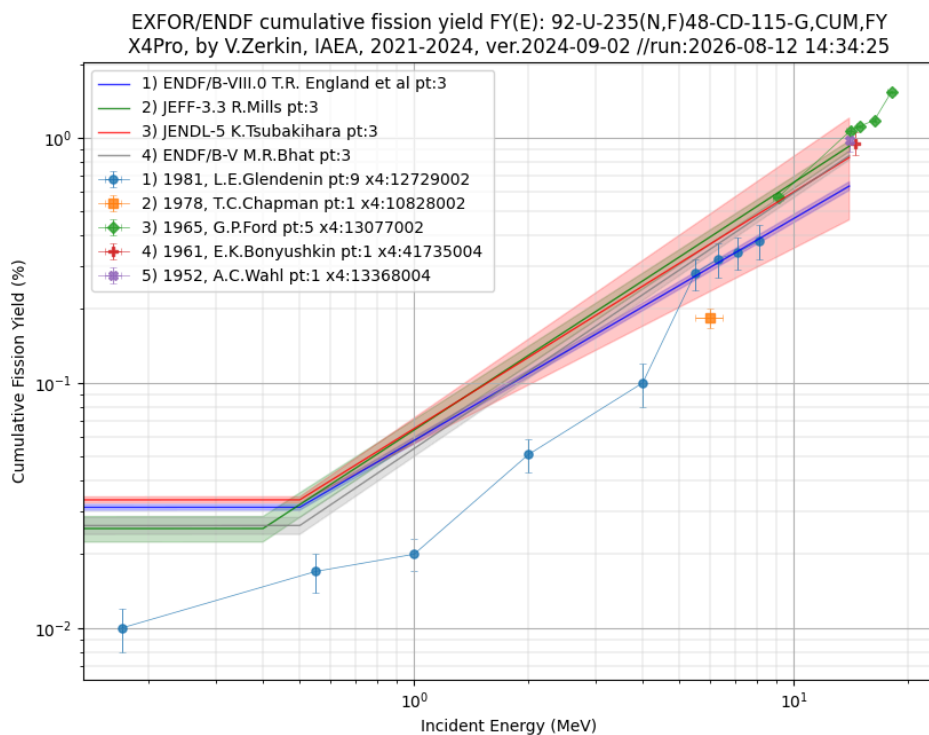


part2-7-fye/ EXFOR/ENDF fission yeild (energy distribution)

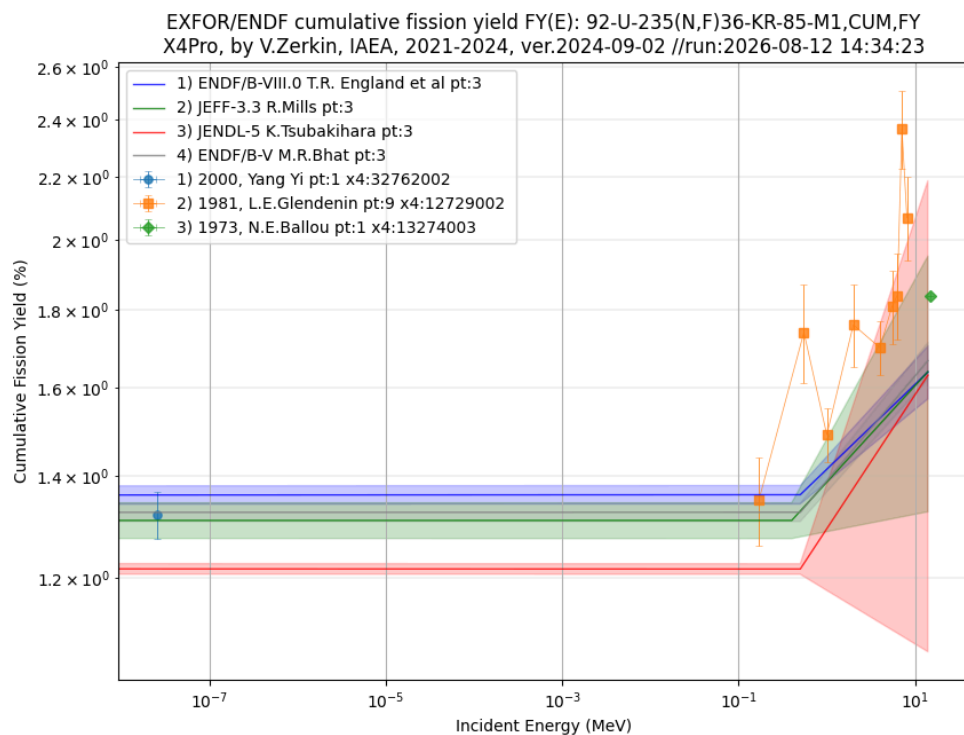
21) part2-7-fye/out00/Pu239-nf-Mo99-cumFY.png



22) part2-7-fye/out00/u235-nf-cd115g-cumFY.png

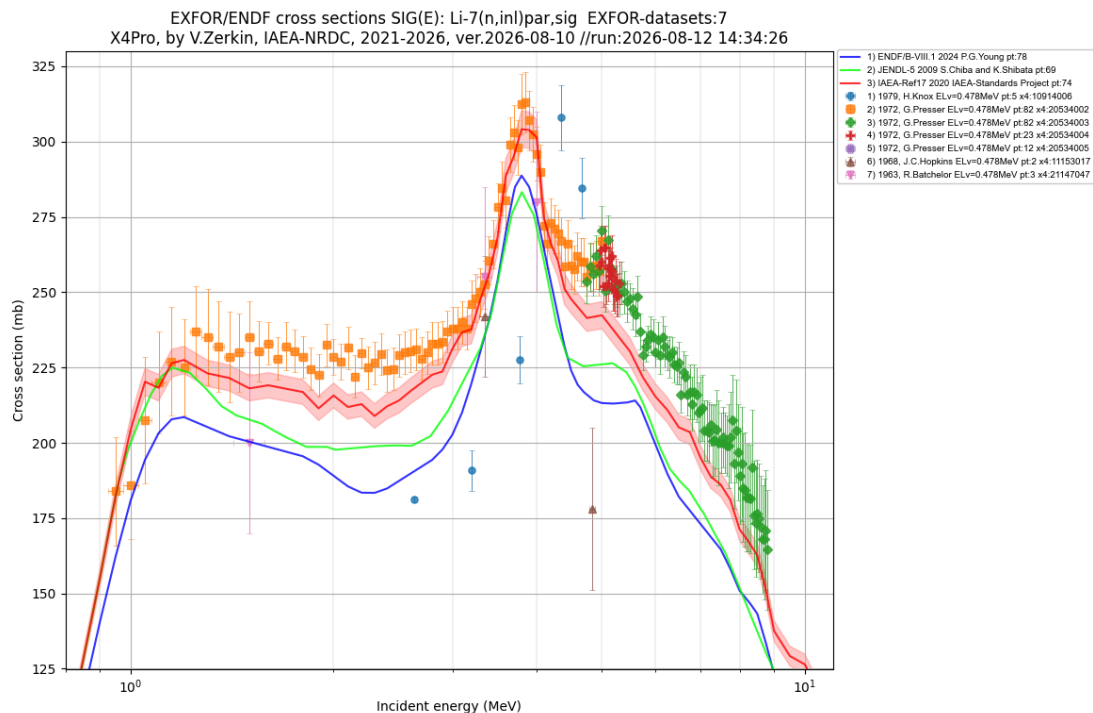


23) part2-7-fye/out00/u235-nf-kr85m-cumFY.png



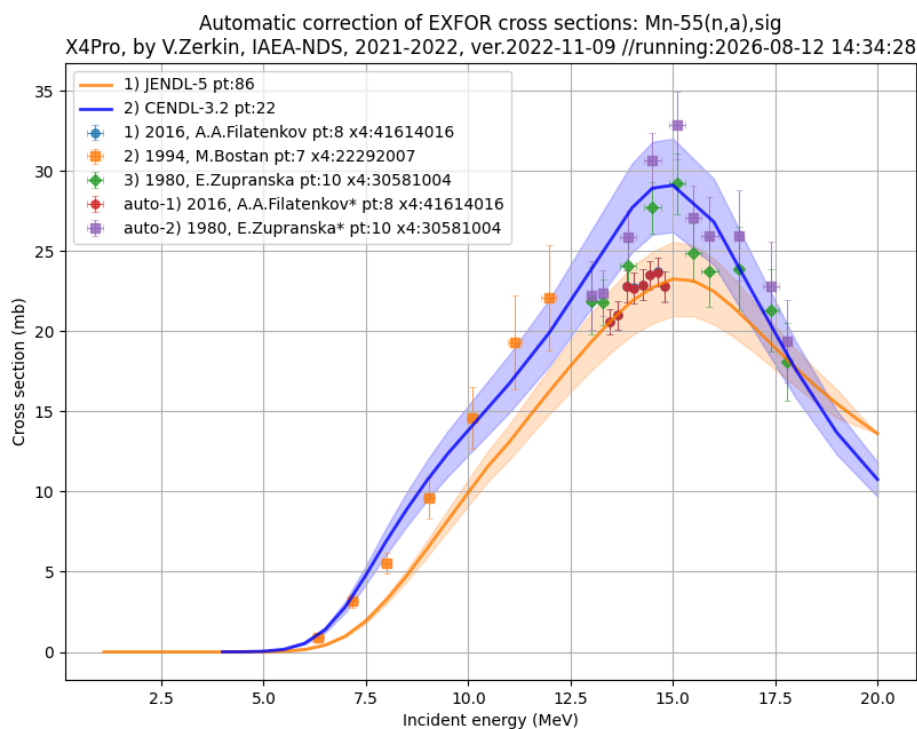
part2-8-sig1par/ EXFOR/ENDF partial cross sections CSP:MT51

24) part2-8-sig1par/out00/Li7ninl.png



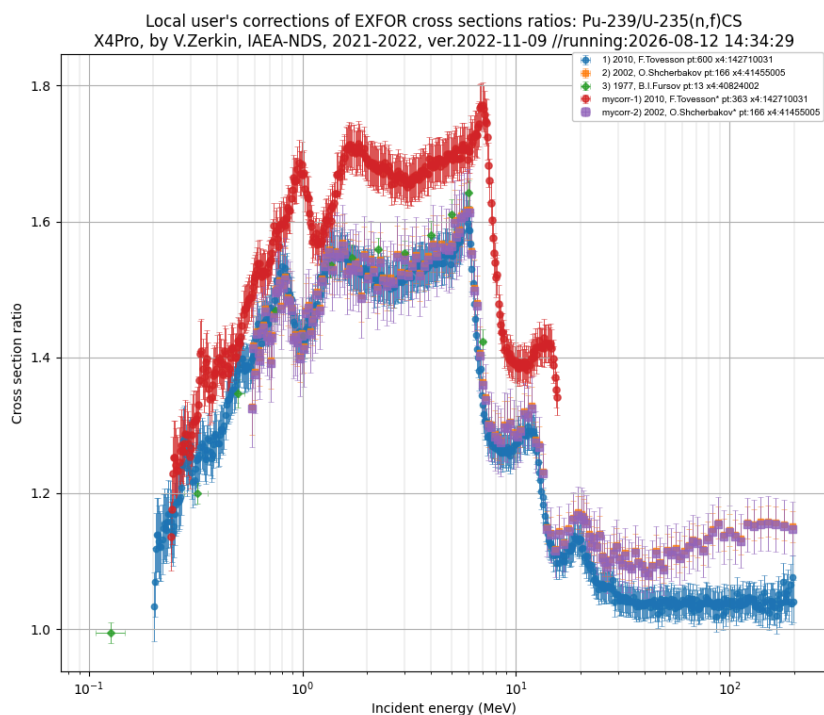
part3-1-auto1/ Automatically renormalize EXFOR CS using old monitor CS and new standards

25) part3-1-auto1/out00/auto1.png



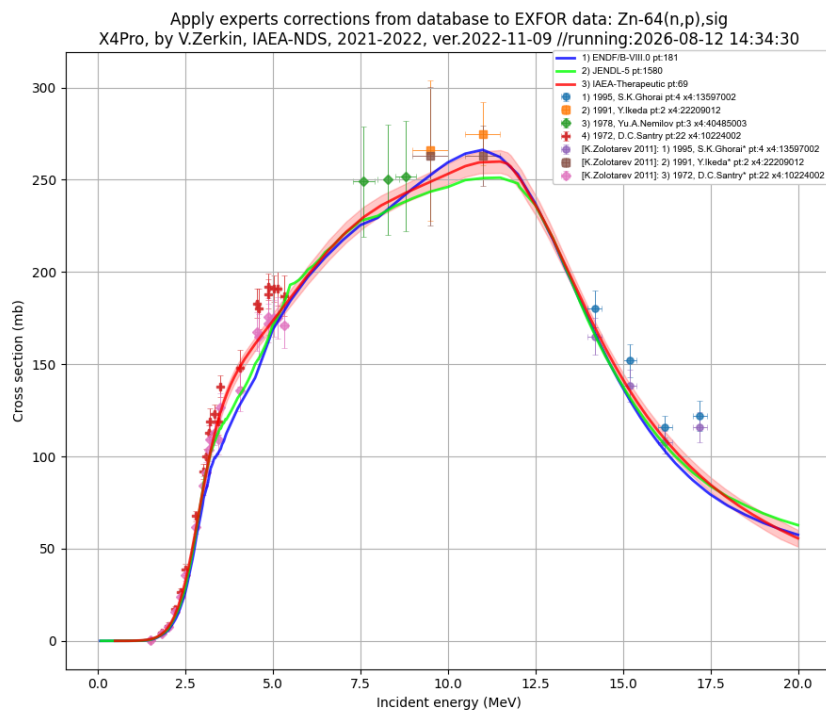
part3-2-user1/ Correct data from EXFOR by user's program

26) part3-2-user1/out00/user1.png



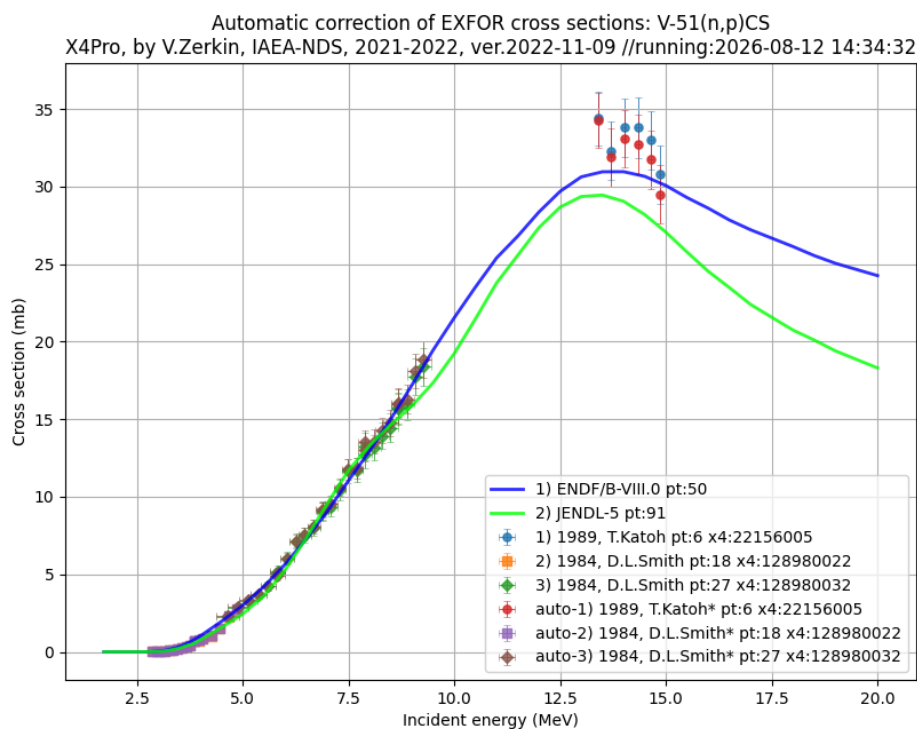
part3-3-expert1/ Apply experts corrections stored in X4Pro database

27) part3-3-expert1/out00/expert1.png



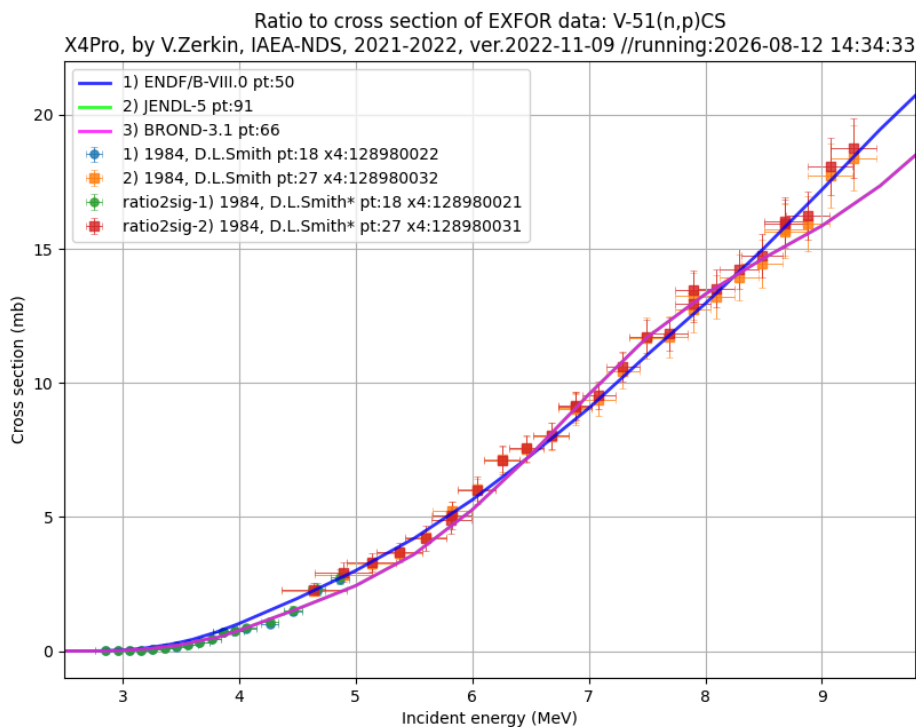
part3-4-auto2/ Automatically renormalize EXFOR CS using old monitor CS and new standards, old and new decay data

28) part3-4-auto2/out00/auto2.png



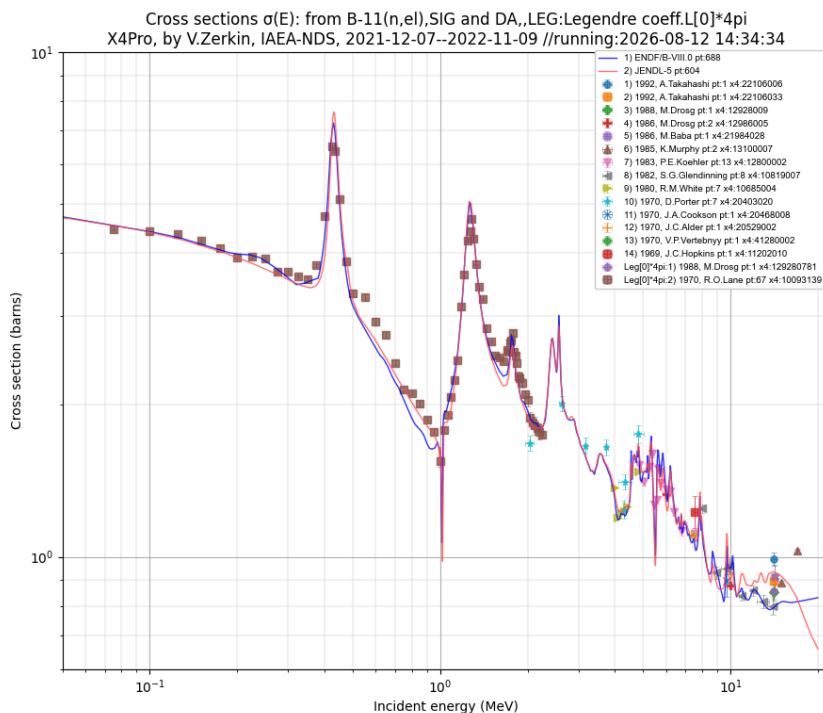
part3-5-ratio2sig/ Retrieve ratios, calculate CS and uncertainties using latest standards

29) part3-5-ratio2sig/out00/ratio2cs.png



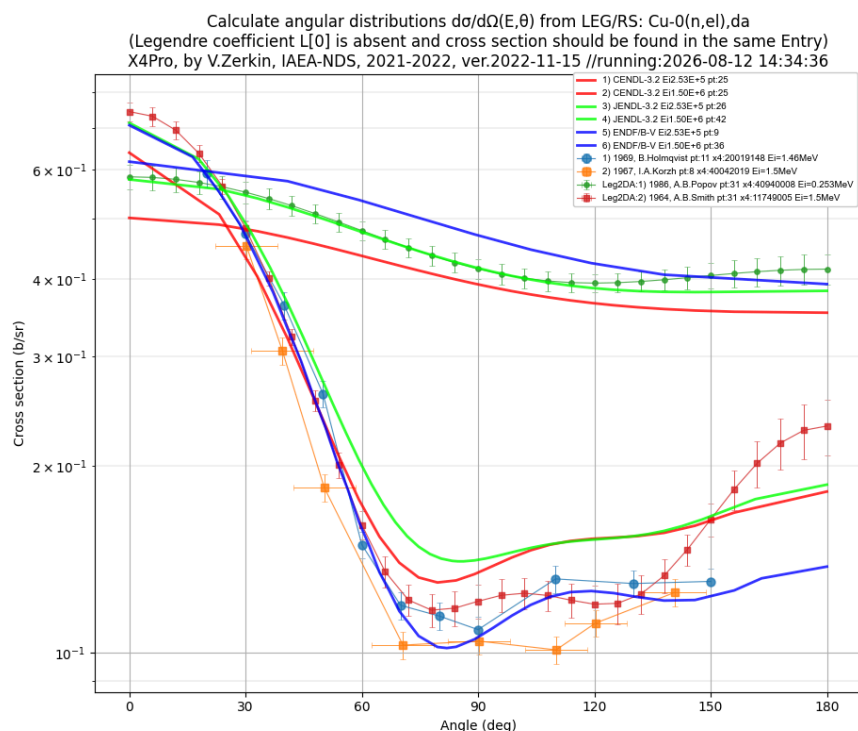
part3-6-daleg2sig/ Retrieve from EXFOR: Legendre coefficient $L[0]$, calculate cross sections and errors

30) part3-6-daleg2sig/out00/B11nel.png



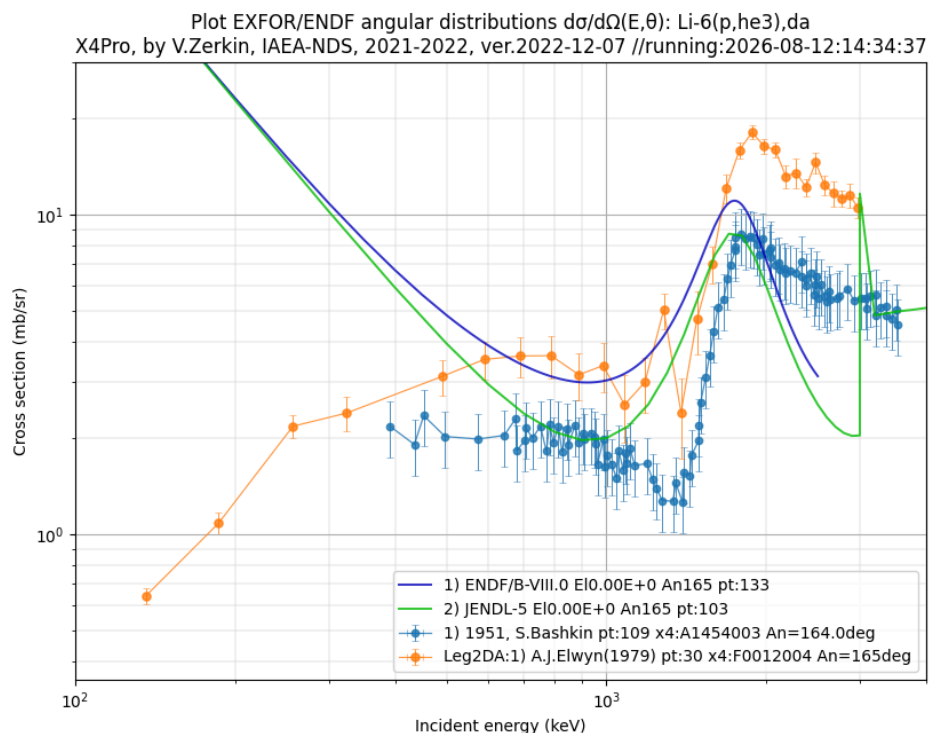
part3-7-legrs2da/ EXFOR LEG/RS and SIG from the same Entry; calc. missing Legendre coefficient $L[0]$; calc. angular distributions

31) part3-7-legrs2da/out00/Cu0nel_da.png



part3-8-leg2r33/ Retrieve Legendre coefficient calculate angular distributions store R33dat (draft of R33)

32) part3-8-leg2r33/out00/da1ei_leg2da.png



part4-0-couchdb/

33) part4-0-couchdb/out1/10010.png

The screenshot shows the CouchDB web interface in a browser. The address bar indicates the URL is `localhost:5984/_utils/#database/zv-exfor-001/10010`. The interface has a dark sidebar on the left with navigation links: Databases, Setup, Active Tasks, Configuration, Replication, News, Documentation, Verify, and Your Account. The main content area displays the document for key `10010` in the `zv-exfor-001` database. At the top of the document view, there are buttons for `Save Changes`, `Cancel`, `Upload Attachment`, `Clone Document`, and `Delete`. The document is a JSON object with the following structure:

```
1 {
2   "_id": "10010",
3   "_rev": "1-2b049342859403e09e912b1bb46c50b2",
4   "ENTRY": "10010",
5   "compiled": 20050707,
6   "x4dbVersion": "2023-03-15",
7   "year": 1970,
8   "a1": "A.B.SMITH",
9   "ref": "",
10  "title": "Fast-neutron total and scattering cross sections of bismuth.",
11  "TransID": "1336",
12  "TransDate": 20050926,
13  "INSTITUTE": [
14    {
15      "x4pointer": " ",
16      "x4codes": [
17        {
18          "code": "IUSAAML",
19          "dict": "INSTITUTE",
20          "idict": 3,
21          "hlp": "Argonne National Laboratory, Argonne, IL, United States of America"
22        },
23        {
24          "code": "3SAFPEL",
25          "dict": "INSTITUTE",
26          "idict": 3,
27          "hlp": "Atomic Energy Corp.of South Africa, Pelindaba, South Africa, Rep."
28        }
29      ]
30    }
31  ],
32  "REFERENCE": [
33    {
34      "x4pointer": " ",
35      "x4codes": [
36        {
37          "code": "J,NSE,41,63,197007",
38          "stdFileName": "J,NSE,41,63,1970",
39          "year": 1970,
40          "typ": "J",
41          "ref": "J,NSE",
42          "vol": "41",
43          "p": "63",
44          "shortRef": "Jour: Nuclear Science and Engineering, Vol.41, p.63 (1970)"
45        }
46      ]
47    }
48  ],
49 }
```

At the bottom left of the interface, it says "CouchDB" and "Fauxton on CouchDB v 3.2.2". There are also links for "Log Out" and "exfor".

34) part4-0-couchdb/out1/_all_docs.png

The screenshot shows the Project Fauxton web interface for a CouchDB database named 'zv-exfor-001'. The browser address bar shows 'localhost:5984/_utils/#database/zv-exfor-001/_all_docs'. The interface includes a sidebar with navigation options: All Documents, Run A Query with Mango, Permissions, Changes, and Design Documents. The main area displays a table of documents with columns 'id', 'key', and 'value'. The table lists 12 documents, each with a checkbox, a document icon, and a truncated JSON value. At the bottom, it shows 'Showing document 1 - 12.' and 'Documents per page: 100'.

	id	key	value
☐	10010	10010	{ "rev": "1-2b049...
☐	10031	10031	{ "rev": "1-548c6...
☐	10093	10093	{ "rev": "1-685e2...
☐	10116	10116	{ "rev": "1-87a57...
☐	10186	10186	{ "rev": "1-2904af...
☐	10224	10224	{ "rev": "1-ba7db...
☐	10417	10417	{ "rev": "1-a4c93...
☐	10685	10685	{ "rev": "1-3a6d1...
☐	10819	10819	{ "rev": "1-6df51e...
☐	10859	10859	{ "rev": "1-cafade...
☐	11010	11010	{ "rev": "1-4ebb7...
☐	11028	11028	{ "rev": "1-54150...

35) part4-0-couchdb/out1/_find.png

Project Fauxton - database/zv-ex

localhost:5984/_utils/#database/zv-exfor-001/_find

Bing OASIS Search Retiring in... Home - Praxisplan WPEC-SG50 International Atomi...

zv-exfor-001 > Mango Query

Query history

Mango Query

```

1 {
2   "selector": {
3     "year": {
4       "$gte": 1970
5     },
6     "TransID": {
7       "$gt": "0000"
8     }
9   },
10  "fields": [
11    "ENTRY",
12    "year",
13    "a1",
14    "title",
15    "TransDate"
16  ]
17 }

```

Run Query Explain manage indexes

Executed in 12 ms

Table {} JSON

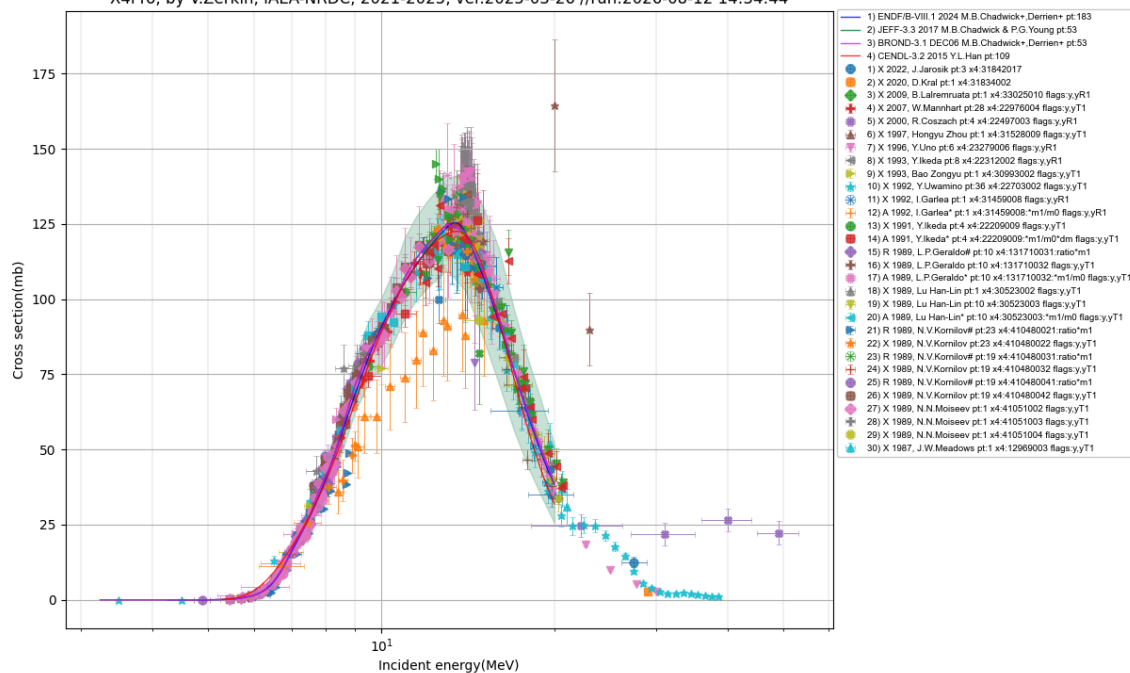
ENTRY	TransDate	year	a1
10010	20050926	1970	A.B.SMITH
10186	20220214	1971	G.N.Salaita
10685	20081211	1980	R.M.White

Showing 5 columns. ☐ Show all columns. Document

part6-1-eval2sig/ Retrieve CS, ratios, monit-data; renormalize to new standards and decay data; convert ratios to CS

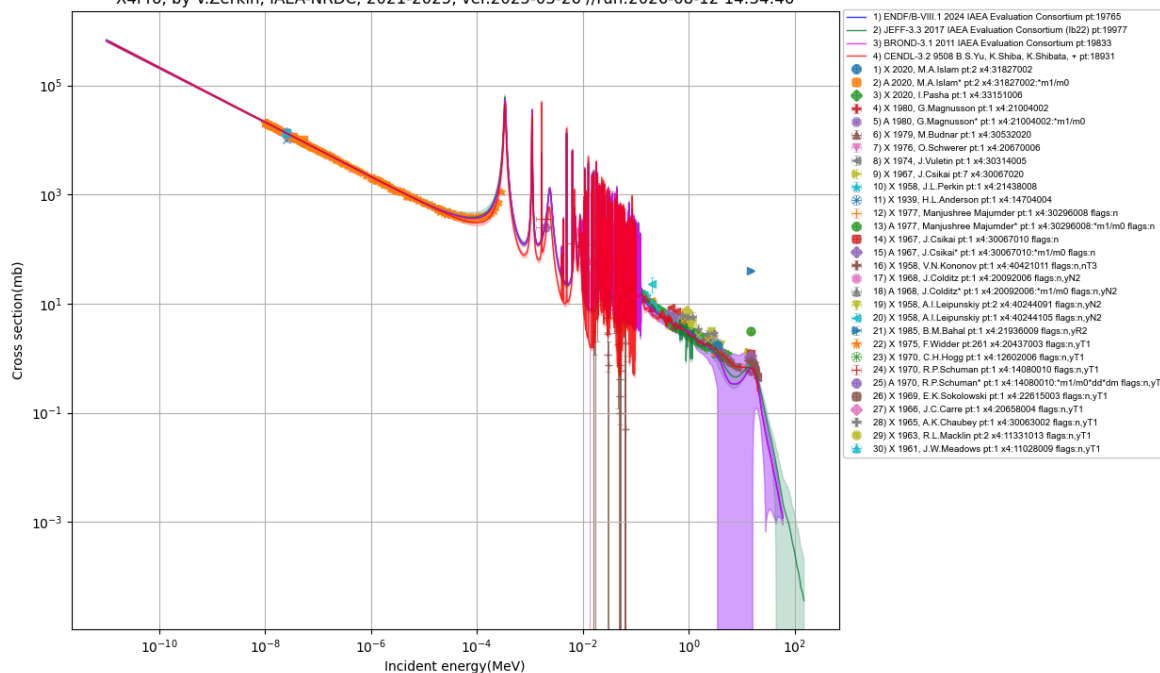
36) part6-1-eval2sig/out00/Al27na.png

Cross sections $\sigma(E)$: Al-27(n,a),sig datasets:116 original:85X auto-corrected:22A* ratio2cs:9R#
 Eva-flags: y:accepted, n:rejected, R:reviewed, T:theoretical comparison, N:no pdf, 1:good, 2:doubtful, 3:outlier
 X4Pro, by V.Zerkin, IAEA-NRDC, 2021-2025, ver.2025-03-26 //run:2026-08-12 14:34:44



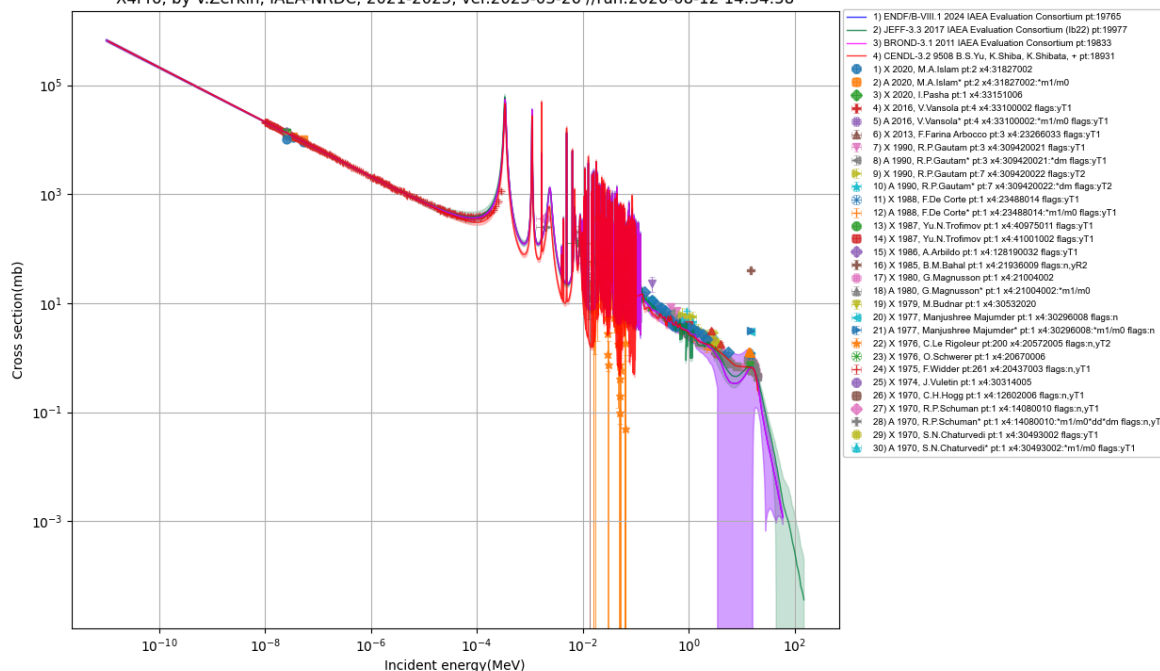
37) part6-1-eval2sig/out00/Mn55ng-n.png

Cross sections $\sigma(E)$: Mn-55(n,g),sig datasets:61 original:49X auto-corrected:12A* ratio2cs:0R#
 Eva-flags: y:accepted, n:rejected, R:reviewed, T:theoretical comparison, N:no pdf, 1:good, 2:doubtful, 3:outlier
 X4Pro, by V.Zerkin, IAEA-NRDC, 2021-2025, ver.2025-03-26 //run:2026-08-12 14:34:46



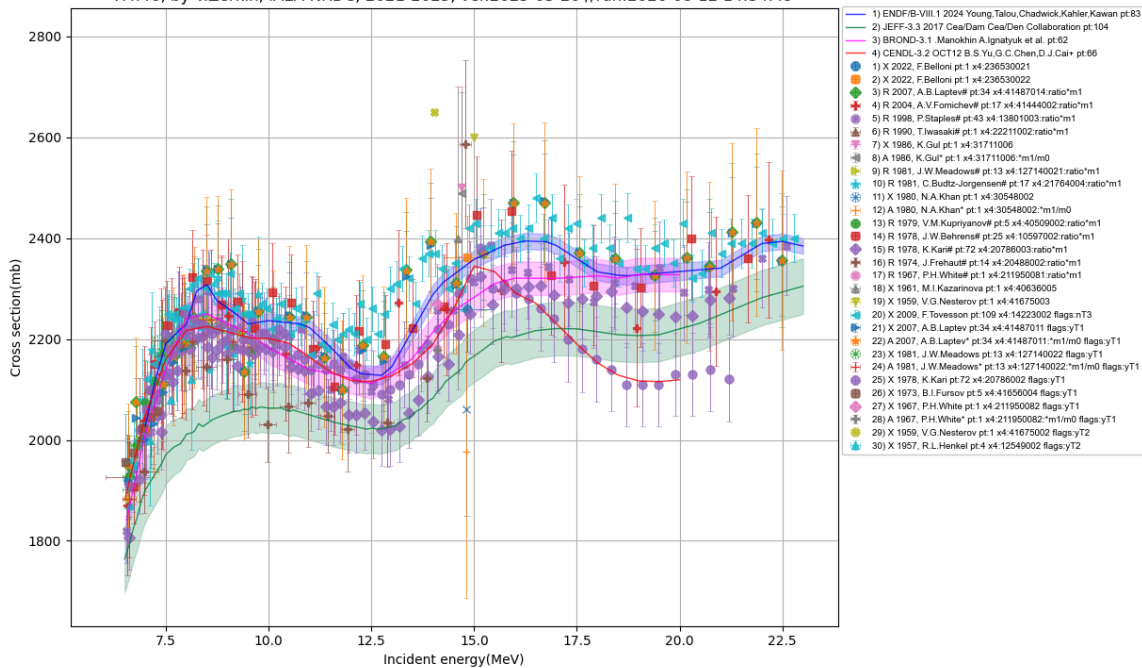
38) part6-1-eval2sig/out00/Mn55ng.png

Cross sections $\sigma(E)$: Mn-55(n,g),sig datasets:61 original:49X auto-corrected:12A* ratio2cs:0R#
 Eva-flags: y:accepted, n:rejected, R:reviewed, T:theoretical comparison, N:no pdf, 1:good, 2:doubtful, 3:outlier
 X4Pro, by V.Zerkin, IAEA-NRDC, 2021-2025, ver.2025-03-26 //run:2026-08-12 14:34:58



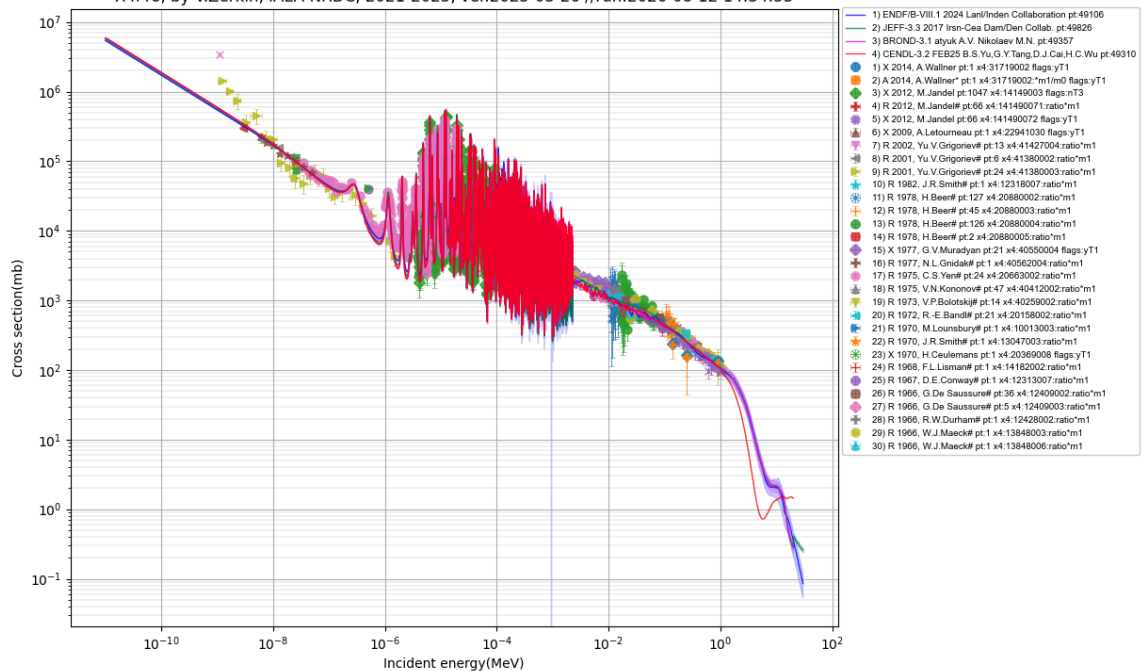
39) part6-1-eval2sig/out00/Pu240nf-n.png

Cross sections $\sigma(E)$: Pu-240(n,f),sig datasets:30 original:14X auto-corrected:5A* ratio2cs:11R#
 Eva-flags: y:accepted, n:rejected, R:reviewed, T:theoretical comparison, N:no pdf, 1:good, 2:doubtful, 3:outlier
 X4Pro, by V.Zerkin, IAEA-NRDC, 2021-2025, ver.2025-03-26 //run:2026-08-12 14:34:49



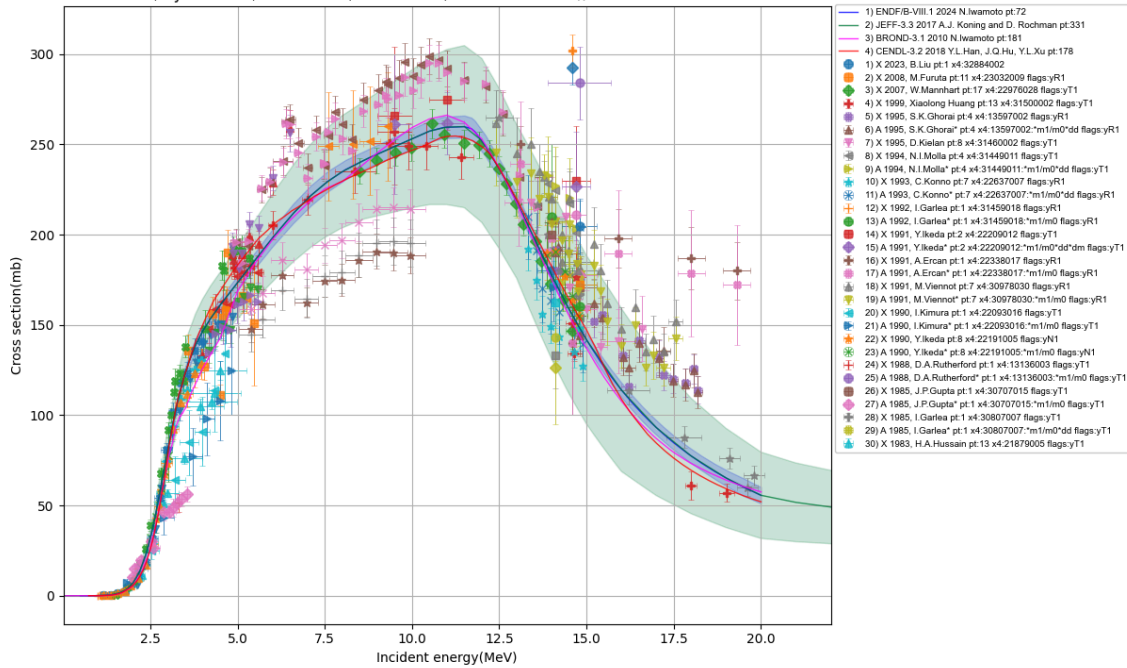
40) part6-1-eval2sig/out00/U235ng.png

Cross sections $\sigma(E)$: U-235(n,g),sig datasets:53 original:10X auto-corrected:1A* ratio2cs:42R#
 Eva-flags: y:accepted, n:rejected, R:reviewed, T:theoretical comparison, N:no pdf, 1:good, 2:doubtful, 3:outlier
 X4Pro, by V.Zerkin, IAEA-NRDC, 2021-2025, ver.2025-03-26 //run:2026-08-12 14:34:53



41) part6-1-eval2sig/out00/Zn64np.png

Cross sections $\sigma(E)$: Zn-64(n,p), sig datasets:87 original:51X auto-corrected:30A* ratio2cs:6R#
 Eva-flags: y:accepted, n:rejected, R:reviewed, T:theoretical comparison, N:no pdf, 1:good, 2:doubtful, 3:outlier
 X4Pro, by V.Zerkin, IAEA-NRDC, 2021-2025, ver.2025-03-26 //run:2026-08-12 14:34:51



Created by [V.Zerkin](#), 09-Aug-2026
 Last updated: 08/16/2026 22:49:22